

Highlights

- Quantitative-coupled BowTie for high-rpm valve-train design assessment
- Top event: loss of valve–cam follower contact (float, bounce, coil bind)
- Indicators: dynamic ratio, natural frequency, coil-bind margin, Goodman FoS
- Two concurrent at-risk barriers identified in the baseline AP 1.8 head
- Qualitative validation against four published valve-train failure cases